

# Final Project Assignment

## Exercise 1 - Baseline Synthesis (2 points)

Using the original `synthesis.tcl` script provided in `WORK_Group26.zip`, synthesize the given RTL design using the STcmos65 HVT library.

**Note:** The files `tech/synopsys_dc.setup` and `tech/synopsys_pt.setup` are not provided and must be implemented.

Find the minimum clock period, among values that are multiples of 0.5 ns, such that the following constraints are met, with both metrics measured in PrimeTime:

- Slack  $\geq 0$
- Area  $< 24100$

**Note:** the `synthesis.tcl` script cannot be modified.

For the selected solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Clock period (ns)
- Area (um2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation.

## Exercise 2 - Power Optimization (2 points)

### 2.1 Clock Gating

Starting from the clock period found in Exercise 1, synthesize the design with clock gating enabled.

Apply clock gating with the following settings:

- Minimum bit-width: 8
- Maximum fanout: 32

For the selected solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Area (um2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation. In the report, discuss and motivate the impact of clock gating on PPA metrics.

## 2.2 Clock Gating and Multi-VT Synthesis

Starting from the clock period found in Exercise 1, synthesize the design using both:

- Clock gating, with the same settings used in Exercise 2.1
- Multi-VT synthesis (with LVT, SVT, and HVT)

**Note:** use same input driving cells, operating conditions and wire load model as in Exercise 1.

For the obtained solution, extend the `pt_analysis.tcl` script to report the following metrics:

- Percentage of LVT cells
- Percentage of SVT cells
- Percentage of HVT cells
- Area (um2)
- Slack (ns)
- Leakage power (mW)
- Dynamic power (mW)

All metrics must be measured in PrimeTime. Power values must be measured using VCD-based SWA back-annotation. In the report, discuss and motivate the impact of Multi-VT synthesis on PPA metrics.

*Suggestion:* In a design with clock gating, use the following command to get the collection of all gates in the design:

```
set gates [get_cells -hierarchical -filter "is_hierarchical == false"]
```

## Exercise 3 - Custom Post-Synthesis Report in PrimeTime (2 points)

Write a TCL procedure named `custom_report` to be executed in Synopsys PrimeTime.

The procedure must accept the following input arguments:

- **num\_paths:** integer value specifying the maximum number of timing paths to analyze;
- **delay\_type:** string specifying the type of timing analysis to perform. Accepted values are:
  - **min:** minimum-delay analysis;
  - **max:** maximum-delay analysis.

The procedure must print a textual report to the PrimeTime command-line interface. For the delay type specified by **delay\_type**, the procedure must consider the **num\_paths** most critical timing paths and report how many times each combinational cell in the design appears in those paths.

Only cells with at least one occurrence in the selected critical paths must be reported.

Each row of the report must have the following format:

```
<cell_full_name> <cell_ref_name> <number_of_occurrences_in_critical_paths>
```

The reported rows must be sorted in descending order according to the number of occurrences in the selected critical paths.

## Deliverables

Submit a zip file named `Group26.zip`.

The archive must contain four folders, one for each exercise, and one final report, following the structure below:

```
Group26/  
  ex1/
```

```

    sha1_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    pt_analysis.tcl
ex2.1/
    sha1_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    synthesis.tcl
    pt_analysis.tcl
ex2.2/
    sha1_core.sdc
    synopsys_dc.setup
    synopsys_pt.setup
    synthesis.tcl
    pt_analysis.tcl
ex3/
    custom_report.tcl
REPORT.pdf

```

Additional requirements:

- In `ex1/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex2.1/synthesis.tcl`, include all clock-gating commands.
- In `ex2.1/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex2.2/synthesis.tcl`, include all clock-gating and Multi-VT synthesis commands.
- In `ex2.2/pt_analysis.tcl`, include the commands for SWA back-annotation and for reporting the requested metrics.
- In `ex3/custom_report.tcl`, include the code of the requested TCL procedure.
- For `REPORT.pdf`, use the tex template available in `WORK_Group26.zip`.

## Suggestions

- Review all lab slides and tutorials.
- Review the solutions to the lab exercises uploaded to the Material page.

## Submission Instructions

- Each group must send an e-mail to `valentino.peluso@polito.it`, with `andrea.calimera@polito.it` in cc.
- Use the following subject: `SODS26 Group26`.
- Attach the file `Group26.zip` containing all required deliverables.

## Deadline: July 7, 23:59

Late submissions, or submissions not compliant with the specifications above, will be automatically discarded.

